Mask-Aware Local–Global Attention for Efficient Image Inpainting

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Abstract

Image inpainting aims to reconstruct missing regions while maintaining consistency with the visible image. Although convolutional methods are effective at modeling local structure, large or semantically complex missing regions often require long-range context. Transformer-based approaches address this with self-attention, but full global attention over high-resolution image tokens is computationally expensive. This project investigates whether global attention is necessary for all tokens during inpainting. This project propose a mask-aware local–global design in which visible regions are processed primarily with efficient local or window-based operations, while masked tokens selectively retrieve long-range information from visible features through global attention. Masked tokens may also interact locally to maintain coherence within the reconstructed region. The proposed approach will be compared with convolutional, local-attention, and full-global-attention baselines using reconstruction quality and computational efficiency across different mask sizes and geometries.

1 Introduction

Image inpainting aims to recover missing or corrupted regions of an image while producing content that is visually and semantically consistent with the observed surroundings. This task is challenging because the evidence needed to fill a missing region may lie far beyond its immediate neighborhood.. For example, completing a large missing object or structured scene may require reasoning about distant parts of the image. Early convolutional approaches are effective at modeling local texture and structure, but their local receptive fields can make explicit long-range information transfer difficult. Methods such as contextual attention therefore introduced mechanisms that allow missing regions to directly retrieve relevant features from distant visible regions [1].

More recent inpainting methods have explored architectures with stronger global modeling capabilities. LaMa uses Fourier convolutions to obtain an image-wide receptive field [2], while Transformer-based approaches exploit self-attention to model long-range dependencies. However, applying full global self-attention to high-resolution feature maps is computationally expensive because its cost grows quadratically with the number of image tokens. Existing approaches address this limitation in different ways. For example, MAT introduces mask-aware attention together with window-based processing to aggregate information from valid image regions efficiently [3], while T-former modifies the attention operator itself to reduce its computational complexity [4].

This project investigates a complementary question: *does every image token require global attention during inpainting?* The visible region of an incomplete image is already known and primarily serves as contextual evidence for reconstructing the missing region. Therefore, allowing every visible token to globally query every other token may spend computation on interactions that are not equally important for the reconstruction task. In contrast, masked tokens must infer unknown content and may benefit substantially from direct access to distant visible context. This motivates an asymmetric allocation of local and global computation based on whether a token belongs to the observed or missing region.

The proposed architecture uses mask-aware local–global attention, where visible image features are processed primarily with efficient local or window-based operations, while masked tokens selectively perform global cross-attention over visible features. Masked tokens can additionally interact with nearby masked tokens to maintain spatial consistency within the reconstructed region. The main goal is to test whether masked-to-visible global interactions provide most of the benefit of full global attention. Then compare the proposed design with convolutional, local-attention, and full-global-attention baselines and evaluate both reconstruction quality and computational efficiency across different mask sizes and geometries.

2 Related Work

Early learning-based image inpainting methods were dominated by convolutional architectures. Yu et al. [1] introduced contextual attention to allow missing regions to explicitly retrieve relevant features from distant visible regions, addressing the difficulty of standard convolutions in modeling long-range dependencies. Liu et al. [5] proposed partial convolutions, in which convolutional responses are conditioned only on valid pixels and the validity mask is progressively updated through the network. More recently, LaMa [2] used Fast Fourier Convolutions to obtain an image-wide receptive field, showing that strong global context can also be achieved without self-attention.

Transformer-based methods further explored explicit long-

range reasoning for image completion. TFill [6] combines restrictive local convolutional tokenization with Transformer-based global context modeling, separating local feature extraction from long-range reasoning. MAT [3] introduces a mask-aware Transformer for large-hole inpainting, where a dynamic mask prevents invalid features from contributing to attention and shifted-window processing enables efficient information propagation. CMT [7] extends this idea by replacing binary validity with a continuous mask that represents different levels of token reliability during feature propagation. These methods demonstrate the importance of both mask awareness and long-range interactions for recovering large or structured missing regions.

A separate line of work focuses on reducing the computational cost of attention. T-former [4] develops a linear-complexity attention formulation for image inpainting rather than applying standard quadratic self-attention. In video inpainting, ProPainter [8] introduces mask-guided sparse attention that discards unnecessary query and key/value tokens to improve efficiency. Unlike methods that make attention cheaper throughout the network, this project restricts global querying to masked tokens and studies how this choice affects reconstruction quality and computational cost.

3 Proposed Method

Let x denote the original image and M a binary mask, where $M_i = 1$ indicates a missing region. The observed input is

$$x_{\text{obs}} = (1 - M) \odot x. \tag{1}$$

The image is divided into patches and embedded as tokens. Visible patches are represented using their image content, while missing patches are replaced by a learned mask embedding. Positional embeddings are added so that different masked locations remain distinguishable even though they share the same mask representation.

Our model first applies local or window-based attention to the token grid. This stage captures nearby texture and structure efficiently and also allows neighboring masked tokens to exchange information. This introduces a selective global-attention stage in which only masked tokens act as queries, while visible tokens provide keys and values. Let H_M and H_V denote the masked and visible token features, respectively. The global contextual update is

$$\hat{H}_M = \text{Softmax}\left(\frac{Q_M K_V^\top}{\sqrt{d}}\right) V_V, \tag{2}$$

where $Q_M = H_M W_Q$, $K_V = H_V W_K$, and $V_V = H_V W_V$. Thus, every missing location can directly retrieve relevant information from anywhere in the observed image without requiring all visible tokens to perform global queries.

The updated masked representations are decoded to predict the missing image content. Since visible pixels are already known, the final reconstruction preserves them directly:

$$\hat{x}_{\text{final}} = (1 - M) \odot x + M \odot \hat{x}. \tag{3}$$

For N total tokens and K masked tokens, full global self-attention requires approximately $O(N^2)$ pairwise interactions. The proposed global stage instead requires approximately $O(K(N - K))$, in addition to the cost of local window attention. This suggests a potential efficiency advantage when the masked region is relatively small or moderate. The project will therefore explicitly study how this tradeoff changes as the mask ratio increases.

4 Experimental Plan

Experiments will target 256×256 images, with 128×128 used for initial debugging and architecture validation if needed. Both rectangular and irregular masks will be used at 10%, 25%, 50%, and 75% masking ratios to evaluate how selective global attention performs as the missing region increases.

Four main models will be compared: (1) a convolutional U-Net baseline, (2) a model using only local/window attention, (3) a model using full global self-attention, and (4) the proposed mask-aware local–global model. Where feasible, a MAT-style mask-aware window-attention baseline will also be included. All models will be trained under similar settings to ensure a controlled comparison.

The initial training objective will use an ℓ_1 reconstruction loss over the masked region,

$$\mathcal{L}_{\text{rec}} = \frac{1}{|M|} \sum_{i \in M} |x_i - \hat{x}_i|. \tag{4}$$

If necessary, a perceptual loss may be added to improve visual quality. Reconstruction quality will be evaluated using PSNR, SSIM, and LPIPS, with masked-region results reported separately so that unchanged visible pixels do not dominate the evaluation. In addition, ablation studies will remove the global masked-to-visible attention, vary the number of global-attention blocks, and evaluate different mask sizes and geometries. The main objective is to determine whether selective global attention can preserve the reconstruction quality of full global attention while reducing computational cost.

5 Task Assignment

This project will be completed individually. I will be responsible for the literature review, method design, implementation, model training, experimental evaluation, ablation studies, result analysis, and preparation of the final report and presentation.

References

[1] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. S. Huang, "Generative image inpainting with contextual attention," in *CVPR*, 2018.

[2] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, and V. Lempitsky, "Resolution-robust large mask inpainting with fourier convolutions," in *WACV*, 2022.

[3] W. Li, Z. Lin, K. Zhou, L. Qi, Y. Wang, and J. Jia, "MAT: Mask-aware transformer for large hole image inpainting," in *CVPR*, 2022.

[4] Y. Deng, S. Hui, S. Zhou, D. Meng, and J. Wang, "T-former: An efficient transformer for image inpainting," in *ACM Multimedia*, pp. 6559–6568, 2022.

[5] G. Liu, F. A. Reda, K. J. Shih, T.-C. Wang, A. Tao, and B. Catanzaro, "Image inpainting for irregular holes using partial convolutions," in *ECCV*, 2018.

[6] C. Zheng, T.-J. Cham, J. Cai, and D. Phung, "Bridging global context interactions for high-fidelity image completion," in *CVPR*, 2022.

[7] K. Ko and C.-S. Kim, "Continuously masked transformer for image inpainting," in *ICCV*, 2023.

[8] S. Zhou, C. Li, K. C. K. Chan, and C. C. Loy, "ProPainter: Improving propagation and transformer for video inpainting," in *ICCV*, 2023.